READ ONCE · SPEAK OFF-HAND

Canton & Sotto, in plain language.

Everything you need to narrate the demo without notes — what Canton is, what Sotto does, how it works end to end, and crisp answers to the questions a judge is most likely to ask.

> **The one-breath pitch:** Sotto is a confidential payout rail on Canton. A company pays its whole team in one batch — but each person sees only their own payment, an auditor can verify the entire batch, and it all settles atomically. Privacy isn't bolted on by the app; it's a property of the ledger.

1 What is Canton? (say this if asked)

Canton is a **privacy-enabled, public Layer-1 blockchain** built for regulated finance. It's the network the demo settles on, created by Digital Asset and powered by the **Daml** smart-contract language. The single most important thing to understand — and the thing that makes Sotto possible:

> THE LINE THAT MATTERS
>
> On most blockchains, every node sees every transaction — privacy is an add-on. On Canton it's the reverse: **data is only ever shared with the parties who are party to it.** There is no global public ledger everyone can read. That's called *sub-transaction privacy*, and it's enforced by the protocol, not the app.

The five ideas you should be able to say

TERM	IN ONE SENTENCE
Party	An actor on the ledger — a company, a person, an auditor. Like an account.
Participant node	The server that hosts parties and stores *only* the contracts its parties are involved in — never the whole chain.
Daml contract	The smart contract. It names its **signatories** (who must authorise + can see it) and **observers** (who can see it). That declaration *is* the privacy.
Sub-transaction privacy	The ledger discloses a contract only to its stakeholders. Two parties in the same transaction can be blind to each other's parts.
Atomic settlement	One transaction can touch many parties and either fully commits or fully aborts — all-or-nothing — even across parties who can't see each other.

> WHY THIS COMBINATION IS RARE
>
> Lots of systems are private. Lots are atomic. Canton is **private *and* atomic at the same time** — many parties settle in one indivisible transaction while each sees only their slice. That exact combination is what a confidential payout needs, and it's hard to get anywhere else.

2 The problem Sotto solves

Every blockchain payment is public by default. So if a company pays its whole team in one on-chain transaction, it has just **published the payroll** — every employee can see what every other employee earns, and so can competitors and the public.

That single fact is why payroll, grants, supplier runs, and token distributions still happen off-chain, or leak. **Privacy is the thing keeping real money off public rails.** Sotto removes that blocker.

3 What Sotto is — one batch, four kinds of eyes

A payout isn't one audience — it's four. Sotto serves all four from **one shared batch**, and the ledger projects the correct view to each:

LENS	WHO, IN THE DEMO	SEES
Payer	Lumen Studio	The whole batch — every name, role, amount. Runs the payout.
Recipient	Amara Okafor	**Only her own line.** The other five never reach her.
Approver	Priya Raman	Only the one payment over the threshold, to co-sign.
Auditor	Hale & Co.	Every receipt — read-only, by grant.

> **Say it precisely:** the recipient's hidden lines aren't *blurred* or *redacted* on her screen — they **never arrive**. The ledger never disclosed them to her. She can't even learn how many other people were paid.

The four contracts that make it true

Privacy "falls out of" how four Daml contracts declare who's a stakeholder. You don't need to recite this, but it's the proof under the hood:

CONTRACT	WHO CAN SEE IT	WHAT IT GUARANTEES
`Holding`	issuer + owner	A balance is visible only to its owner. Recipients never see each other's money.
`DisbursementReceipt`	recipient + auditor	Each recipient sees *only their own* payment; the auditor sees *all*. This asymmetry is the demo.
`PayoutMandate`	payer + auditor	The on-ledger spending authority — cap, threshold, allow-list. Its `Disburse` choice settles the batch atomically.
`LargePaymentProposal`	+ approver	Maker-checker: an over-threshold payment is held until the approver signs, on-ledger.

4 How it works, end to end

The story you can walk through from memory:

1. **Log in.** A user signs in with email / SSO — no crypto onboarding (see §5).
2. **A wallet appears.** The backend provisions their Canton party just-in-time. No seed phrase, no extension.
3. **The payer sets a mandate.** Spending authority on the ledger: a cap, an approval threshold, the recipient allow-list, and the named auditor.
4. **The payer drops in a batch** — here, six contributors totalling 52,550.
5. **Disburse.** Five under-threshold lines settle in **one atomic transaction** — all or nothing. The treasury drops.
6. **The big one is held.** Kwame's 32,000 is over the 25,000 threshold, so it waits for the approver's signature — maker-checker, on-ledger.
7. **Receipts are written.** Each recipient gets a receipt only they and the auditor can read.

8. **The auditor verifies** the whole batch, read-only — because the payer named them on the mandate, never assumed.

5 No wallets to "connect" (a quiet advantage)

Canton is **not** MetaMask. There's no browser extension, no seed phrase, no "Connect Wallet" button. Identity is a signed login token (JWT, RS256 over JWKS); the token maps to a Canton party.

- **Recipients & staff** — log in with email, get paid, cash out. The participant node custodies their identity. Zero key management.
- **The payer / treasury** — can self-custody via Canton's external-signing, but that's enterprise key management (HSM / signer service), never a consumer popup.

> QUOTABLE
> "No seed phrases, no wallet extensions. People log in like any app and get paid. The crypto is invisible — the privacy is the only thing you notice."

6 What's real vs. seeded (be honest — it reads as strength)

Real

- The four Daml contracts; privacy *asserted* green in a test script.
- A backend over Canton's JSON Ledger API — RS256/JWKS auth, JIT wallets, real `Holding` reads/writes.
- The privacy & atomic-settlement *mechanism*.

Seeded

- The figures in the web console (treasury, the six amounts).
- Done deliberately so the demo never depends on a live chain being up.
- The data layer is shaped to swap seeds for the live API behind the same interface.

> SAY THIS IN THE CLOSE
> "The web console runs on sample data so the demo is rock-solid. The confidentiality and settlement it shows are real — proven by the Canton model and backend in this same repo."

7 The numbers (so you never fumble one)

THING	VALUE
Batch	Lumen Studio · 6 contributors
Batch total	`52,550 USDCx`
Approval threshold	`25,000 per payment`
Over-threshold line	`Kwame Nyong · 32,000 (held for approval)`
Recipient lens follows	`Amara Okafor · 4,200`
Treasury seed (payer)	`312,480 → 291,930 (after 5 settle) → 259,930 (if Kwame approved)`
Auditor / Approver	Hale & Co. (read-only) · Priya Raman (second signer)

USDCx is a stand-in for a tokenized deposit / stablecoin — the settlement token, represented by the `Holding` contract.

8 Why Canton and not Ethereum (the differentiator)

	PUBLIC CHAIN (E.G. ETHEREUM)	CANTON
Default	Everything public; privacy bolted on (mixers, ZK).	Private by default; disclosure is opt-in per stakeholder.
Multi-party settle	Atomic — but everyone sees it.	Atomic *and* confidential at once.
Fit for payroll	Leaks the whole list.	Each party sees only their line; an auditor sees all.

> IF ASKED "WHY NOT ETHEREUM + ZERO-KNOWLEDGE?"
> "ZK adds privacy on top of a chain that's public by default — a lot of complexity to claw back what Canton gives you natively. Canton was built for regulated finance: private by default, atomic multi-party settlement, and selective disclosure to an auditor. Sotto is downstream of that design, not fighting it."

9 Judge Q&A bank (your off-hand insurance)

Q Isn't a permissioned/private chain not really a blockchain?

Canton is a *public, shared* network with privacy — not a private silo. Parties across different organisations transact on one interoperable ledger with real finality; they just don't broadcast their data to everyone. Permissioning is at the party level, not a walled garden.

Q How is the privacy actually enforced — what stops me reading another line?

The Daml contract names its stakeholders (signatories/observers). The protocol only delivers a contract to those parties. The recipient's API call physically can't return someone else's line because that data was never disclosed to her participant node. It's protocol-level, not an app-level filter.

Q Does "atomic" mean everyone can see the transaction?

No — that's the clever part. Parties who can't see each other's lines still settle in one indivisible transaction. Atomicity without disclosure.

Q What stops the payer hiding payments from the auditor?

The auditor is an observer on the mandate *and* on every receipt by construction. The mandate — its cap, threshold, allow-list — is on the ledger. The payer can't disburse outside it without the auditor seeing.

Q Is the money real?

The token is a `Holding` contract — USDCx, standing in for a tokenized deposit. In the web demo the figures are seeded so it's reliable; the read/write mechanism is real on the Canton backend in the repo.

Q Why not just use a private database?

Because no single party is trusted to hold the truth. Canton gives mutually-distrusting parties one cryptographically-consistent ledger where each sees only their slice, settles atomically, and an auditor can verify — a database run by one company can't offer the others that guarantee.

Q How do recipients onboard — do they need a wallet?

They log in with email/SSO. The backend provisions their Canton party just-in-time — no extension, no seed phrase. (See §5.)

Q What's the maker-checker / approver for?

Any payment over the mandate threshold is held until a second party co-signs, on-ledger. The approver sees *only* the over-threshold line — not the routine payments, which settled without them.

Q Does it scale?

Canton scales horizontally — add participant nodes and synchronizers. Because each node only processes data for its own parties, privacy and scale push the same direction rather than fighting.

Q What's the business — who pays?

First wedge: confidential payroll and contractor payouts for orgs that want stablecoin rails but can't expose comp. Then supplier settlement, grants, and tokenized-deposit payouts — anywhere many private payments must settle together and stay auditable. Payments & neobanking primary; RWA & tokenized deposits secondary.

10 Lines that land (sprinkle these)

- "One batch. Four kinds of eyes."
- "Privacy here isn't a feature the app bolts on — it's a property of the rail."
- "The other lines aren't hidden from her. They never arrived."
- "Atomic *and* confidential — settle together, see only your own part."
- "Visibility is granted on purpose, never assumed — and never lets you move funds."
- "The crypto is invisible. The privacy is the only thing you notice."

11 Glossary (one line each)

Canton	Privacy-enabled public L1 blockchain for regulated finance; runs Daml.
Daml	The smart-contract language; contracts declare signatories & observers.
Party	An actor on the ledger (company, person, auditor).
Participant node	Hosts parties; stores only the contracts they're stakeholders in.
Synchronizer	Orders & commits transactions; gives finality (formerly "domain").
Signatory	Party that must authorise a contract — and can see it.
Observer	Party that can see a contract without authorising it.
Sub-transaction privacy	Each party sees only the parts of a transaction it's a stakeholder in.
Atomic settlement	A multi-party transaction fully commits or fully aborts.
Holding	The settlement-token contract (a balance). Sotto's USDCx.
Mandate	On-ledger spending authority: cap, threshold, allow-list, auditor.
Maker-checker	Two-party control: large payments need a second signer.
JWKS / RS256	How login tokens are signed & verified to authorise a party.